

Total No. of Questions : 8]

SEAT No. :

PD-4887

[Total No. of Pages : 2

[6404]-444

B.E. (AI & ML)

**HONORS IN MACHINE LEARNING
(2019 Pattern) (Semester - VII) (410301)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What is Tree based method? Describe following tree based methods. [9]

- i) Regression Trees
- ii) Classification Trees

b) Explain SVM as a Penalization Method in detail. [8]

OR

Q2) a) Elaborate out of Bag Samples, Variable Importance, Proximity Plots w.r.t random forest in detail. [9]

b) Define SVM. Explain in detail Support Vector Machines for Regression. [8]

Q3) a) Write a short note on : [9]

- i) Recurrent Networks
- ii) Regularization in Neural Networks

b) What is learning time? Explain Time Delay Neural Networks and Recurrent Networks in detail. [9]

OR

P.T.O.

- Q4) a)** Explain following backpropagation algorithm in multilayer perceptron in detail. [9]
- Two-Class Discrimination
 - Multiclass Discrimination
- b)** Why Tuning the Network Size required in multilayer perceptron? How it done? [9]

- Q5) a)** Explain in detail Gaussian Mixtures as Soft K-means Clustering. [9]
- b)** Consider the market basket transaction shown below. [8]

Transaction ID	Item Bought
T1	Mango, Apple, Banana, Dates
T2	Apple, Dates, Coconut, Banana, Fig
T3	Apple, Coconut, Banana, Fig
T4	Apple, Banana, Dates

Assuming the support of 50% and confidence of 80%

- Find all frequent itemsets using Apriori algorithm
- Find all association rules using Apriori algorithm

OR

- Q6) a)** Explain with suitable example k-medoids algorithm. [8]
- b)** Write short note on : [9]
- Market Basket Analysis
 - Vector Quantization

- Q7) a)** Write short note on : [9]
- Markov Random Fields
 - Learning the Structure of a Graphical Model
- b)** Explain Model Selection in HMM in detail. [9]

OR

- Q8) a)** Explain how to find state sequence and model parameters in HMM in detail. [9]
- b)** What do you mean by linear regression? Explain it with suitable example. State which applications are best modeled by linear regression. [9]
